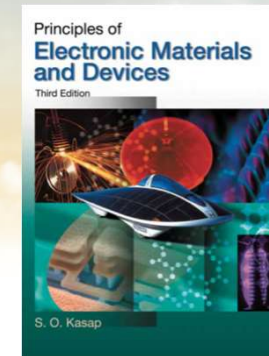




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch3-2 Elementary Quantum Physics

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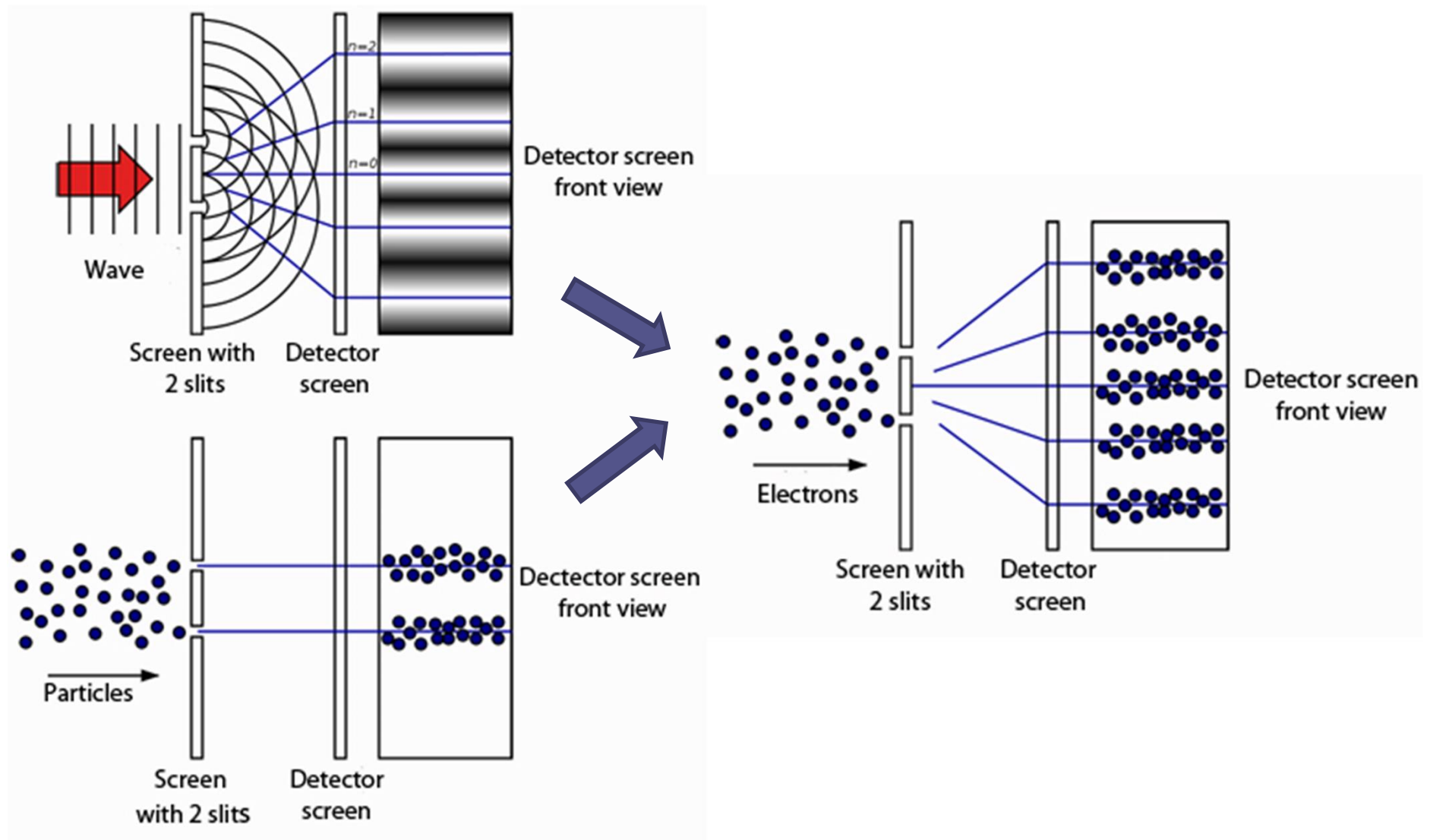
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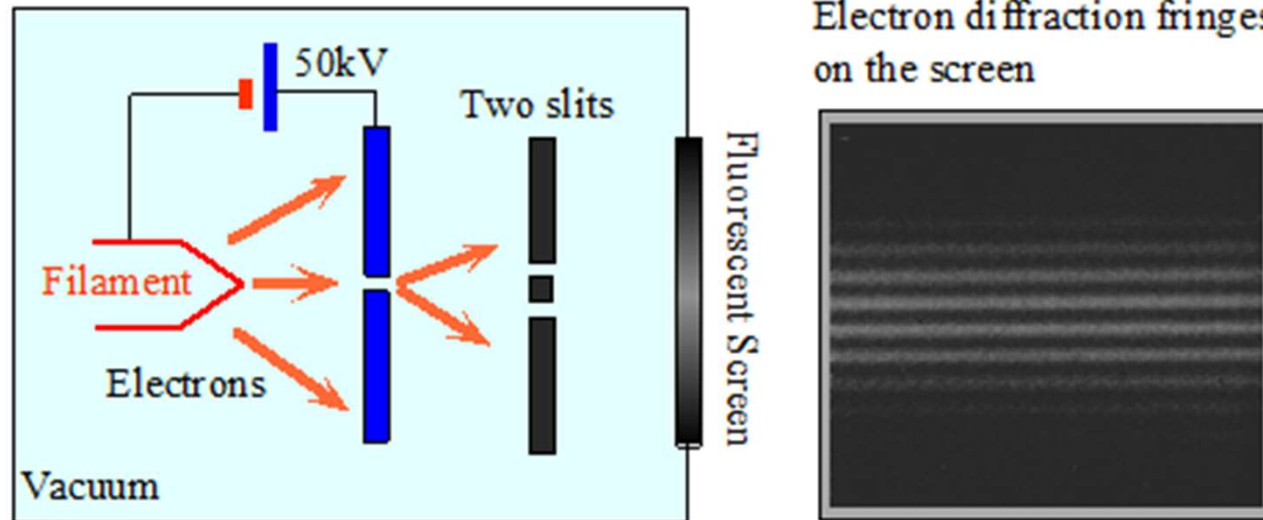


Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

The Electron as a Wave



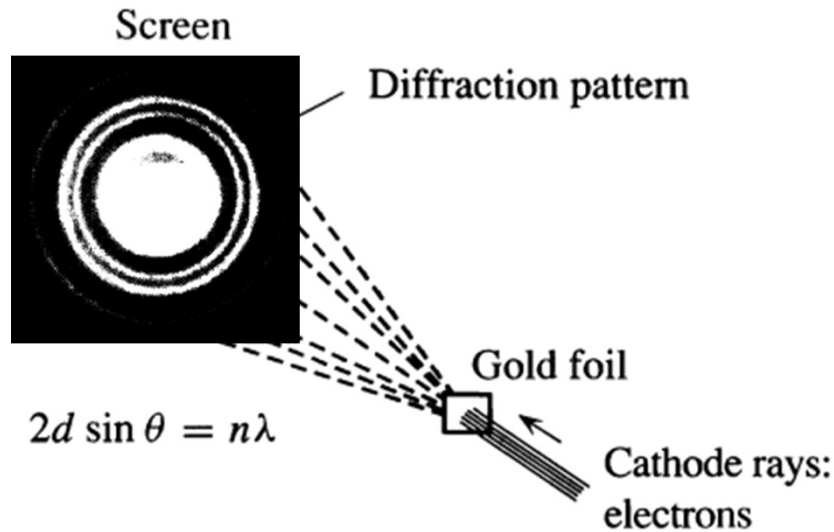
The Electron as a Wave



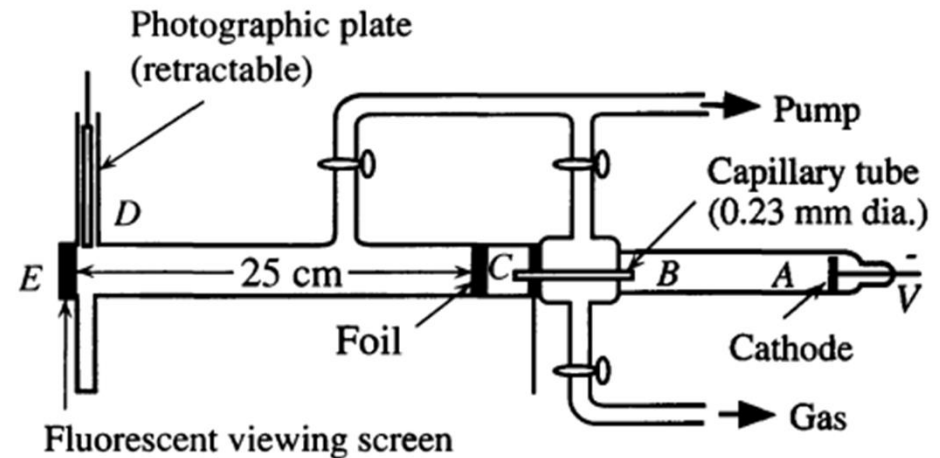
- ❑ The hypothesis of de Broglie (1924)
All moving objects have wave as well as particle characteristics
- ❑ An e travelling with a momentum p behaves like a wave of wavelength λ ,
given by $\lambda = \frac{h}{p}$ (de Broglie wavelength)
- ❑ Provide the basis on which we use wave theory to describe the motion and behavior of electron in a crystal.



The Electron as a Wave



(a) Thomson diffracted electrons by using a thin gold foil and produced a diffraction pattern on the screen of his apparatus in (b). The foil was polycrystalline, so the diffraction pattern was circular rings.



(b) In Thomson's electron diffraction apparatus a beam of electrons is generated in tube A, passed through collimating tube B, and made to impinge on a thin gold foil C. The transmitted electrons impinge on the fluorescent screen E, or a photographic plate D, which could be lowered into the path. The entire apparatus was evacuated during the experiment.

Since we know the interatomic spacing d and we can measure the angle of diffraction 2θ , we can readily evaluate the wavelength λ associated with the wave-like behavior of the electrons. Furthermore, from the accelerating voltage V in the electron tube, we can also determine the momentum of the electrons, because the kinetic energy gained by the electrons, $(p^2/2m_e)$, is equal to eV . Simply by adjusting the accelerating voltage V , we can therefore study how the wavelength of the electron depends on the momentum.

$$\lambda = \frac{h}{p}$$



Examples

- ① A 50g golf ball traveling at a velocity of 20 m/s, the wavelength is

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{Js}}{50 \times 10^{-3} \text{kg} \times 20 \text{m/s}} = 6.63 \times 10^{-34} \text{m}$$

The h is so small while the m is large for macroscopic objects, leading to a very small wavelength. No interference can happen!

- ② A proton with $m=1.67 \times 10^{-27} \text{ kg}$ travelling at 2200 m/s, the wavelength is

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{Js}}{1.67 \times 10^{-27} \text{kg} \times 2200 \text{m/s}} = 0.18 \text{ nm}$$

The wavelength is comparable to the lattice of crystal, so interference can happen.



Duality (Wave-Particle Dualism)

<u>popular name</u>	<u>descriptive concept</u>	
	<u>wave</u>	<u>particle</u>
light	Electro magnetic(EM) wave	< photon >
heat	lattice vibration	< phonon >
electron	< de Brogile wave >	electron
* < > : achivement of modern physics		
* photon : particle - like description of EM wave (quantization) $\left(\begin{array}{l} \text{photon eneygy } E = h\nu = \hbar\omega = \frac{hc}{\lambda} \quad (\lambda = c/\nu, \omega = 2\pi\nu) \\ \\ E[\text{ev}] = \frac{1.24}{\lambda[\mu m]} \end{array} \right.$ • de Brogile wave: wave - like description of electron $\left(P = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{p} \quad \text{or} \quad P = \hbar k \quad (k = \frac{2\pi}{\lambda} : \text{wave vector}) \right.$		



de Broglie Matter Wave



The Nobel Prize in Physics 1929

Louis de Broglie

The Nobel Prize in Physics 1929



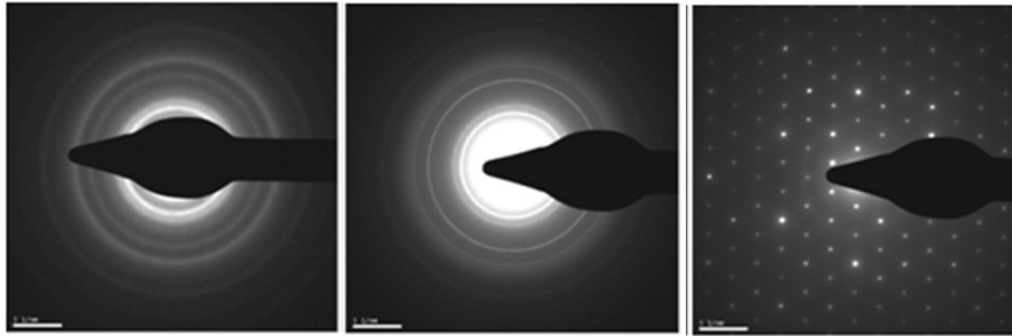
Prince Louis-Victor
Pierre Raymond de
Broglie

The Nobel Prize in Physics 1929 was awarded to Louis de Broglie *"for his discovery of the wave nature of electrons"*.

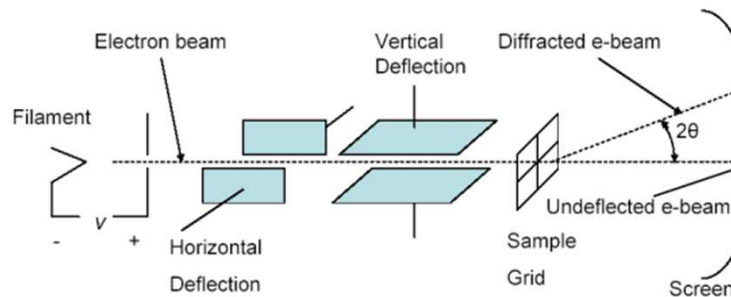


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Electron Diffraction



TEM diffraction patterns for amorphous carbon (Left), polycrystalline aluminum (Center), and single-crystal gold (Right).



$$eV = \frac{1}{2}mv^2 = \frac{p^2}{2m} = E$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \sqrt{\frac{150}{V}} \text{ angstrom}$$

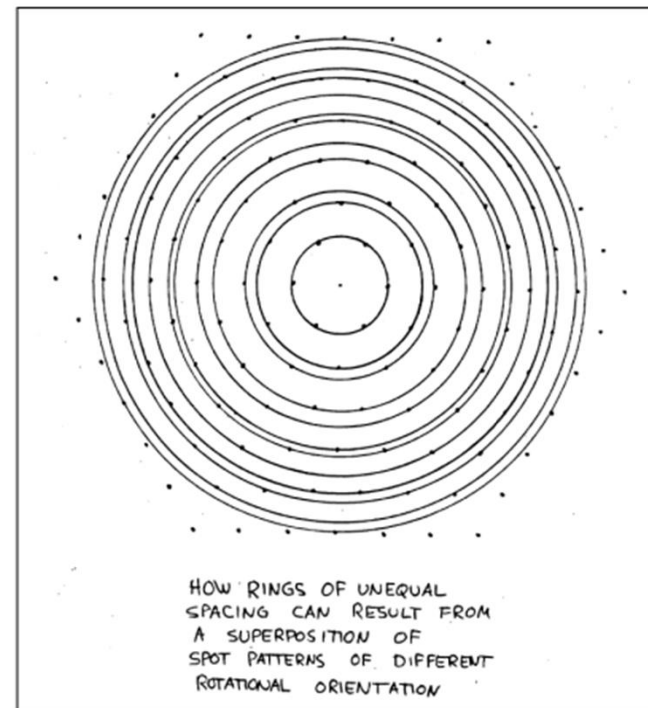


Fig. 6. Spot pattern for a two-dimensional graphite crystal. If the target is composed of many parallel crystals but randomly oriented with respect to an axis through the center of the pattern (non-deflected beam direction) then the different spot patterns superimpose and one gets a "ring" pattern.



Electron Diffraction



The Nobel Prize in Physics 1937

Clinton Davisson, George Paget Thomson

The Nobel Prize in Physics 1937



Clinton Joseph
Davisson



George Paget
Thomson

The Nobel Prize in Physics 1937 was awarded jointly to Clinton Joseph Davisson and George Paget Thomson *"for their experimental discovery of the diffraction of electrons by crystals"*



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Deduction of Wavefunction

- ❑ In 1926, Schrodinger knew of de Broglie's proposal that a moving particle has wavelength, $\lambda = \frac{h}{p}$. With this idea in mind, Schrodinger constructed a differential equation for a wavelike electron.

- ❑ A travelling electromagnetic wave equation:

$$E(x, t) = E_0 \exp j(kx - \omega t) = E_0 \exp(jkx) \exp(j\omega t) = E(x) \exp(-j\omega t)$$

- ❑ Similarly, Schrodinger matter wave function:

$$\psi(x, t) = A \exp j(kx - \omega t) = A \exp j\left(\frac{2\pi}{\lambda}x - 2\pi\nu t\right) = A \exp j\left(\frac{2\pi p}{h}x - \frac{2\pi E}{h}t\right) = A \exp j\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)$$

$$\hbar = \frac{h}{2\pi} : \text{The Dirac constant or the "reduced Planck constant"}$$



Deduction of Wavefunction

$$\psi(x, t) = A \exp\left(j \frac{p}{\hbar} x\right) \exp\left(-j \frac{E}{\hbar} t\right)$$

$$\Rightarrow \frac{\partial \psi}{\partial x} = A j \frac{p}{\hbar} \exp\left(j \frac{p}{\hbar} x\right) \exp\left(-j \frac{E}{\hbar} t\right) = j \frac{p}{\hbar} \psi(x, t)$$

$$\Rightarrow \frac{\partial^2 \psi}{\partial x^2} = -A \frac{p^2}{\hbar^2} \exp\left(j \frac{p}{\hbar} x\right) \exp\left(-j \frac{E}{\hbar} t\right) = -\frac{p^2}{\hbar^2} \psi(x, t)$$

$$\Rightarrow p^2 \psi(x, t) = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}$$

Similarly

$$E \psi(x, t) = -\frac{\hbar}{j} \frac{\partial \psi}{\partial t}$$

$$V \psi(x, t)$$

$$KE + PE = E$$

$$KE \cdot \psi(x, t) = \frac{p^2}{2m} \psi(x, t) = \frac{1}{2m} \left(-\hbar^2 \frac{\partial^2 \psi(x, t)}{\partial x^2} \right)$$

$$PE \cdot \psi(x, t) = V(x) \psi(x, t)$$

$$E \cdot \psi(x, t) = -\frac{\hbar}{j} \frac{\partial \psi(x, t)}{\partial t}$$

Schrödinger's Equation

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x) \psi(x, t) = -\frac{\hbar}{j} \frac{\partial \psi(x, t)}{\partial t}$$



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Schrödinger's Wave Equation

□ One dimension time-independent Schrödinger equation

$$\psi(x,t) = \phi(x)f(t) \quad -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x)\psi(x,t) = -\frac{\hbar}{j} \frac{\partial \psi(x,t)}{\partial t}$$

$$f(t) \left[-\frac{\hbar^2}{2m} \frac{\partial^2 \phi(x)}{\partial x^2} + V(x)\phi(x) \right] = -\frac{\hbar}{j} \phi(x) \frac{df(t)}{dt}$$

Separation variable
分离变量

$$\frac{1}{\phi(x)} \left[-\frac{\hbar^2}{2m} \frac{\partial^2 \phi(x)}{\partial x^2} + V(x)\phi(x) \right] = -\frac{\hbar}{j} \frac{1}{f(t)} \frac{df(t)}{dt} = E$$

$$\begin{cases} -\frac{\hbar}{j} \frac{1}{f(t)} \frac{df(t)}{dt} = E & \Rightarrow f(t) \approx \exp j \left(-\frac{E}{\hbar} t \right) \\ \frac{1}{\phi(x)} \left[-\frac{\hbar^2}{2m} \frac{\partial^2 \phi(x)}{\partial x^2} + V(x)\phi(x) \right] = E & \Rightarrow \boxed{\frac{d^2 \phi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V) \phi(x) = 0} \end{cases}$$



Schrödinger's Wave Equation

- One dimension time-independent Schrodinger equation: $\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi(x) = 0$
- Three dimension time-independent Schrodinger equation: $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$, where $V = V(x, y, z), \psi = \psi(x, y, z)$

Classic Variable	Quantum operator
x	x
$f(x)$	$f(x)$
$p(x)$	$\frac{\hbar}{j} \frac{\partial}{\partial x}$
E	$-\frac{\hbar}{j} \frac{\partial}{\partial t}$

$$K.E + P.E = E$$

$$K.E = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{1}{2m} \left(-\hbar^2 \frac{\partial^2 \Psi(x,t)}{\partial x^2} \right)$$

$$P.E = V(x)\Psi(x,t)$$

$$E = -\frac{\hbar}{j} \frac{\partial \Psi(x,t)}{\partial t}$$

□ General expression

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V(x, y, z, t) \psi = -\frac{\hbar}{j} \frac{\partial \psi}{\partial t}$$

$$\bar{H} \psi(x, y, z, t) = i\hbar \frac{\partial}{\partial t} \psi(x, y, z, t)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

$$\bar{H} = -\frac{\hbar^2}{2m} \nabla^2 + V(x, y, z, t) \quad \bar{H} \psi = E \psi$$

哈密顿量
Hamiltonian



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Schrödinger's Wave Equation

Schrödinger's Equation
$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi(x) = 0$$

$|\Psi(x, y, z, t)|^2$ is the probability of finding the electron per unit volume at x, y, z at time t .

$|\Psi(x, y, z, t)|^2 dx dy dz$ is the probability of finding the electron in a small elemental volume $dx dy dz$ at x, y, z at time t .

Most commonly accepted interpretation due to Max Born

Seems crazy: we never actually know where the particle is, only the probability of finding it there. Even worse – these “probability waves” can interfere constructively/destructively!



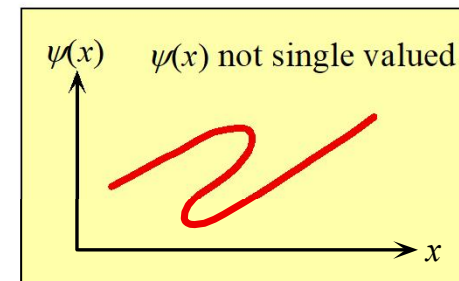
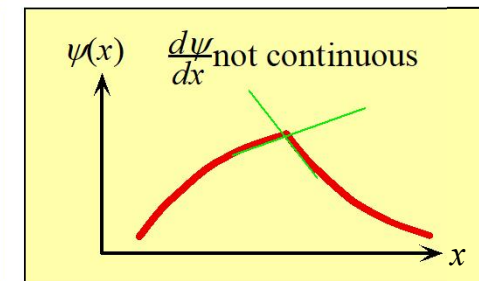
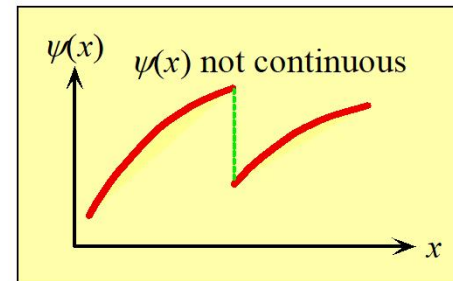
Schrödinger's Wave Equation (Boundary Conditions)

Schrödinger's Equation
$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi(x) = 0$$

- Steady-state probability distribution of the electron is $|\psi(x, y, z, t)|^2 = |\psi(x, y, z)|^2$
- By solving the equation, we will know the probability distribution and the energy of the electron.

□ Boundary conditions:

- ① ψ must be continuous;
- ② $\frac{d\psi}{dx}$ must be continuous;
- ③ ψ is single valued;
- ④ $\int_{-\infty}^{\infty} |\psi(x, y, z)|^2 dx dy dz = 1$



The wave function should satisfy 3 conditions in the whole space: **single value, finite and continuous at every point**. This condition is called the standard condition of the wave function.

波函数在全空间每一点应满足**单值、有限、连续**三个条件。该条件称为波函数的标准条件。

Unacceptable forms of $\psi(x)$



Schrödinger's Wave Equation



The Nobel Prize in Physics 1933
Erwin Schrödinger, Paul A.M. Dirac

The Nobel Prize in Physics 1933



Erwin Schrödinger



Paul Adrien Maurice Dirac

The Nobel Prize in Physics 1933 was awarded jointly to Erwin Schrödinger and Paul Adrien Maurice Dirac *"for the discovery of new productive forms of atomic theory"*



The Nobel Prize in Physics 1954
Max Born, Walther Bothe

The Nobel Prize in Physics 1954



Max Born



Walther Bothe

The Nobel Prize in Physics 1954 was divided equally between Max Born *"for his fundamental research in quantum mechanics, especially for his statistical interpretation of the wavefunction"* and Walther Bothe *"for the coincidence method and his discoveries made therewith"*.



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History of QM

- ① 1900 Plank energy quanta hypothesis
- ② 1905 Einstein photoelectric effect
- ③ 1924 de Broglie matter wave
- ④ 1926 SCH wave equation
- ⑤ 1927 Electron diffraction

